

The surreal-number reports

Seven independent mathematical research packages,
on surreal and surcomplex numbers

September 2026

Scope

This document catalogues the seven research packages stored under `docs`. Each entry names the problem the report attacks, the result it claims, and the archive it arrived in.

These reports were previously part of a larger collection, in a repository whose subject was elsewhere; they were moved here with their git history. That is why an entry may refer to a merge, a duplicate sweep or a delivery number: those describe how the report reached its present form, and the record is kept rather than rewritten.

Each directory holds a typeset article with its $\text{\LaTeX}$ source, a `README.md`, and in most cases verification code together with the recorded output of running it. These are AI-assisted drafts. None is refereed or machine-checked, and this catalogue records what each report claims rather than verifying it.

Rebuild this document with

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latexmk -pdf manifest.tex && latexmk -c manifest.tex
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1 The reports

Six of the seven are about the surreal field itself: birthdays and their behaviour under multiplication, the option graphs of surreal forms, genetic functions and gaps, and two questions of Lipparini about evaluating formal sums in the surreals. The first four of those arrived as two pairs, and each pair was merged.

The seventh, *surcomplex-analysis*, is about the algebraic closure $\mathbf{No}[i]$ rather than $\mathbf{No}$, and is itself a merge of nine manuscripts that attacked one foundational question independently on the same day. It is the longest report here by a wide margin, and its `MERGE_NOTES.md` is worth reading before the article: six of the seventeen load-bearing notions turned out to be inequivalent across those nine sources, so a theorem proved under one is false under another.

1. A square-root obstruction to evaluating increasing Hahn series at ω

[hahn-evaluation-at-omega/article.pdf](#)

from `hahn_evaluation_research.zip` + `reversed_hahn_series.zip`

Answers Problem 7.7 (p. 38) of Lipparini (arXiv:2505.00424v2) negatively: there is no ring homomorphism from the field of set-supported real-coefficient Hahn series with strictly increasing surreal exponents into the surreals sending x^a to ω^a . The certificate is $B \in \mathbb{Q}[[x]]$ with $B^2 = 1 - x$, built by the recursion $b_n = -\frac{1}{2} \sum_{k=1}^{n-1} b_k b_{n-k}$: any such map would force a square to equal $1 - \omega < 0$. The homomorphism is applied only to the finite identity, so no continuity, valuation, order preservation or preservation of infinite sums is assumed. Adds the exact classification of admissible images of the variable as the infinitesimals, the monomial classification over an arbitrary ordered abelian group, and order-uniqueness on the source field. *Merged; the core certificate was identical in both originals and is proved once, but the two toolkits are kept — in particular a nonnegative-coefficient semiring obstruction with strictly weaker hypotheses that the square-root certificate cannot reach.*

2. Gonshor’s Product-Birthday Conjecture for ordinal-support series

[gonshor-product-birthdays/surreal_product_birthdays.pdf](#)

from gonshor_product_birthdays_research.zip

Proves Gonshor's bound $b(xy) \leq b(x) \otimes b(y)$ on the ring of Conway normal forms $\sum_{\alpha \in S} c_\alpha \omega^{-\alpha}$ with S an arbitrary set of ordinals, so transfinite and uncountable supports are allowed. Supplies an exact birthday formula by transfinite induction over the support with maintained prefix bounds and an explicit limit stage, and a complete classification of the equality cases. The route is a leading-Cantor-degree gap, which also gives that the degree of the birthday never rises under multiplication. The conjecture for arbitrary surreal numbers is not solved. The domain is incomparable with that of the Laurent-series report that follows: it excludes ω and every positive power.

3. A Laurent-series case of Gonshor's birthday conjecture

gonshor-laurent-birthdays/article.pdf

from Gonshor_Laurent_Birthday_Research.zip

Proves the same bound $b(xy) \leq b(x) \otimes b(y)$ on the canonically embedded field $\mathbb{R}((\omega^{-1}))$, which unlike the ring of the preceding entry contains ω and its positive powers but restricts supports to order type at most ω on the integer lattice. The route is a bilinear budget: an exact splitting $b(x) = b(P) \oplus b(u)$ into polynomial and infinitesimal parts, three separate product estimates for the resulting cross terms, and a finite-jet stabilisation carrying the finite bound to infinite series without any continuity claim. Also treats finite normal forms with arbitrary ordinal exponents, derives exact birthdays for finite and infinite Laurent series, and proves a reciprocal corollary. Neither this theorem nor the preceding one contains the other; on their common corner $\mathbb{R}[[\omega^{-1}]]$ the two agree and are proved twice.

4. Canonical forms need not be subgraphs

canonical-forms-need-not-be-subgraphs/surreal_graphs.pdf

from surreal_graphs_research.zip + surreal_sparse_graphs.zip

Answers negatively the question in the Figure 3 caption of Roughan, *Surreal Birthdays and Their Arithmetic* (arXiv:1810.10373v2, p. 7): the canonical form of a surreal value need not be a subgraph of every other form of that value. Five vertices and rank three are both least possible, and there are exactly sixteen five-vertex counterexamples, eight triangle-free, four of value $1/2$. Adds cycle representations of $1/2$ at every admissible length, the exact extremal function, a planar construction of maximum degree three and arbitrarily large girth, classifications of which canonical graphs are unavoidable, and transfinite witnesses. *Merged; the two originals exhibited genuinely different five-vertex witnesses — one passing through a vertex of value -1 , one with no negative value at all — and both are printed in full, as are two odd-cycle families, two high-girth constructions and two proofs of the Fibonacci bound.*

5. A countable genetic cut across the finite–infinite gap

genetic-gaps-and-primitives/article.pdf

from genetic_gap_counterexamples.zip

With $q(t) = t/(1+t^2)$ and $B(x) = 1 - \{q(n-x) : n \in \mathbb{N} \mid \}$ as a Conway cut, proves that B is the indicator of the positive infinite surreals, is genetic under the set-sized option constructors of the cited sources, and has pointwise derivative zero everywhere. Consequently x and $x - B(x)$ are genetic primitives of 1 agreeing at every real input that do not differ by a constant, the latter also an increasing bijection. The sublevel class $\{x : 0 < x < \omega, B(x) \leq \frac{1}{2}\}$ is nonempty and bounded yet has no surreal supremum, so neither end gap supplies one.

6. A Number-Valued Broadcast Sum of Surreal Sequences

broadcast-sum-of-surreal-sequences/article.pdf

from surreal_broadcast_sum.zip

Analyses the sign-truncation game of Lipparini (arXiv:2505.00424v2, Remark 7.6(3), in connection with Problem 7.5). Proves the game always has a surreal-number value, that it agrees with finite surreal sums and with Lipparini's ordinal operation on their respective domains, and that it fails weak coordinatewise monotonicity, with an explicit family $s[n] = 3 \cdot 2^{-n-d-1}$ on a subset E and

2^{-n-d} off it realising the failure. The underlying rule is Lipparini's; the name and the notation are local to the report.

7. Surcomplex analysis: holomorphic functions on $\mathbf{No}[i]$

[surcomplex-analysis/article.pdf](#)

from `surcomplex_analysis.zip` and eight companion archives

One article assembled from nine manuscripts, all delivered under the same name on 21 September 2026 as independent attempts at one subject: a theory of holomorphic functions over $\mathbf{K} = \mathbf{No}[i]$ built on Hahn-supported series. The shared core is a calculus with no topological convergence at all. All nine prove that every *set* of surcomplex numbers is closed and discrete, that a convergent set-indexed net is eventually constant, and that every continuous map $[0, 1] \rightarrow \mathbf{K}$ is constant; so $\sum t^n = (1 - t)^{-1}$ is a Hahn identity, never a limit of partial sums, and contour integration is defined coefficientwise on ordinary complex paths. On that base the article develops germs, the coherent algebra $\mathcal{H}(U)$ over an ordinary domain, Weierstrass preparation and infinitesimal zero clusters, a residue calculus, the global principles, exponentials and logarithms, and an all-scale rigidity theorem. **Nine independent runs at a foundation do not agree the way nine runs at a theorem would, and the merge turned on that.** Of seventeen load-bearing notions, six are genuinely inequivalent across the sources, and a theorem proved under one is false under another: there are three inequivalent function classes (the identity theorem, the global maximum principle, Liouville and the open mapping theorem hold for coherent sections and fail for germs; sharp Schwarz and Schwarz–Pick hold for lifts and fail for coherent sections), and three inequivalent residues, two of which disagree numerically — for $1/(\zeta - \varepsilon)$ with ε infinitesimal the cluster residue at the ordinary centre 0 is 1 while the local residue at 0 is 0, the actual pole sitting at ε . A single ordinary bound in Liouville gives a characterization and not constancy: tz is real-bounded by 1 on the whole finite plane and is not constant. Eleven further conflicts are resolved and recorded, including two functions both called “the” logarithm that differ by exactly $2\pi i$. All nine manuscripts and all nine verification programs are kept; the deduplication is in the article.